
You must show work to receive credit. Unless indicated otherwise, each problem is worth equal value.

1. Construct a Venn diagram to determine the validity of the given argument.

- a. 1. All homeless people are unemployed.
2. Bill Gates is not unemployed.

Therefore, Bill Gates is not homeless

- b. 1. No vegetarian owns a gun.
2. All policemen own guns.

Therefore, no policeman is a vegetarian.

- c. 1. No professor is a millionaire.
2. No millionaire is illiterate.

Therefore, no professor is illiterate.

1. Which of the following are statements? Why or why not?

a. $3 + 7 = 9$

b. Solve the equation $3 + 7 = x$.

c. Is $\sqrt{2}$ an irrational number?

2. Write a sentence that negates the statement.

a. Some cats have three legs.

b. No cat has three legs.

c. The cat is not blue.

3. Translate the sentence into symbolic form.

- a.** It is a square and a rectangle.
- b.** If John is male then he is human.
- c.** All squares are rectangles.
- d.** All whole numbers are even or odd.
- e.** No square is a triangle.
- f.** Being an orthodontist is sufficient for being a dentist.
- g.** Not being a monkey is necessary for being an ape.

4. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- a.** All red cars go 140 mph.
- b.** The car is red, goes 140 mph, and does not cost \$70,000.
- c.** If the car does not cost \$70,000, it does not cost \$70,000.
- d.** The car is red, and it does not cost \$70,000 or go 140 mph.
- e.** Being able to 140 mph is sufficient for a car to cost \$70,000 or be red.
- f.** Not being red is necessary for a car to cost \$70,000 and not go 140 mph.